

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20 % of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	S.MYTHILI	Reg. No.:	1924 11250
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 – 50 Q2 – 50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Trace the logical flow from a recurrence relation to its final complexity using the substitution method. Add suitable directional links to show how guessing and induction support the proof of the derived complexity. Justify the role of verification in confirming the correctness of the assumed solution. Interpret how this map provides a stepwise framework for solving recurrence relations systematically.

Q2. Map the relationships among different cases and explain how they represent algorithm behavior. Extend the map by linking asymptotic notation to each case. Justify the importance of analyzing all three cases. Interpret how this map helps in realistic performance comparison.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplar	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

		comparisons			
Clarity	10 %	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20 %)	Integration of Literature (20 %)	Clarity (10 %)	Total Marks (Each – 50)
--------------	------------------------------	----------------	---------------------	----------------------------------	----------------	-------------------------

Q1

Q2

Overall Total (100)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Concept Map: From Recurrence Relation to Final Complexity using Substitution Method:

1. Recurrence Relation
 $T(n) = aT(n/b) + f(n)$
gives the time for
problem size n

2. Guess the solution
guess asymptotic form
 $T(n) = O(g(n))$
based on expansion.

3. Induction Hypothesis
assume for all $m < n$,
 $T(m) \leq c \cdot g(m)$,
where $c > 0$ is a
constant

4. Substitute into
Recurrence
Substitute assumed
bound into recurrence
 $T(n) = aT(n/b) + f(n)$

Final complexity
Therefore;
 $T(n) = O(g(n))$ is
The final time complexity

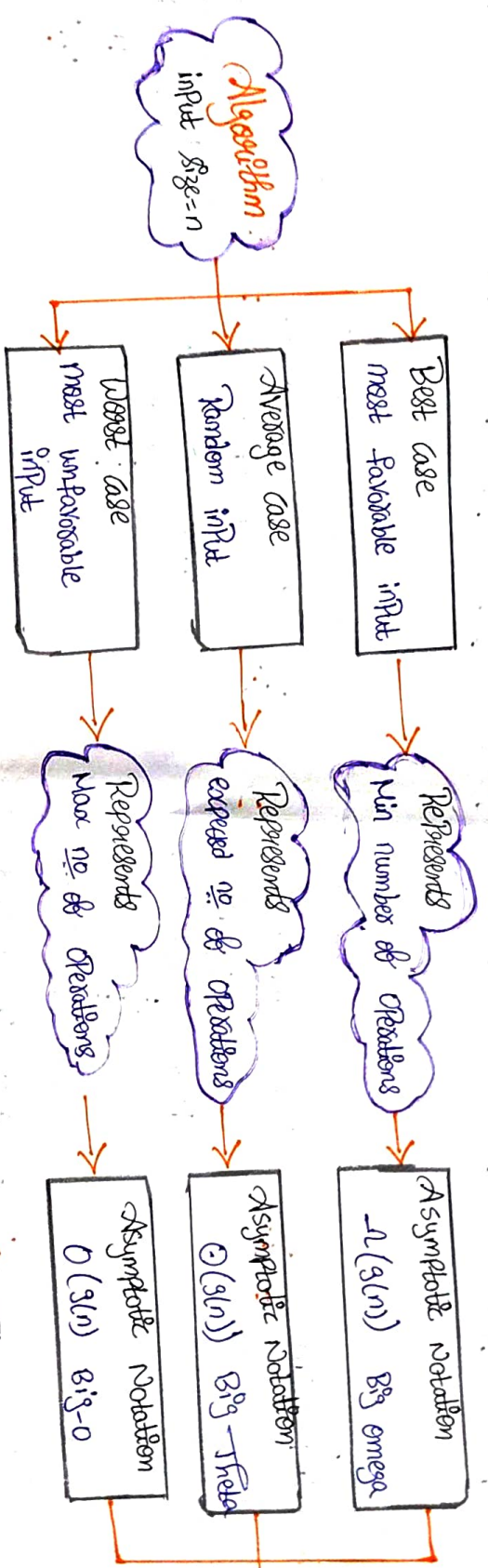
Role of Guessing
* Provides a candidate
bound for solution
* Based on intuition,
Pattern matching.

Role of Induction
* Assumes the solution
holds for smaller
subproblems
* Proves it must also
hold for size n

5. Simplify & Show
Bound
Algebraic Manipulation
to show $T(n) \leq c \cdot g(n)$
for all $n \geq n_0$

Role of Verification
* ensures the assumed solution
is valid for starting point
* ensures the proof is correct
for all n
* Prevents wrong conclusions

2. Relationship Among different cases & asymptotic notation:



Why all 3 cases are important?

- * best case shows lower limit of time
- * worst case provides upper limit
- * Average case shows typical performance

Practical Performance comparison

- * Comparing algorithms using all 3 cases gives a complete picture
- * useful in choosing the right algorithm for specific applications
- * Prevents misleading conclusions based on only one case.